

LAG SELECTION CRITERIA

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VAR Order Selection (* highlights the minimums)

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	AIC	BIC	FPE	HQIC
0	-6.103	-6.100	0.002236	-6.102
1	-11.18	-11.16	1.400e-05	-11.17
2	-11.23	-11.21	1.325e-05	-11.22
3	-11.24	-11.21	1.317e-05	-11.23
4	-11.25	-11.21	1.297e-05	-11.24
5	-11.26	-11.21	1.282e-05	-11.25
6	-11.28	-11.22*	1.265e-05	-11.26
7	-11.28	-11.21	1.265e-05	-11.25
8	-11.29	-11.21	1.250e-05	-11.26*
9	-11.29*	-11.20	1.246e-05*	-11.26

PROCESSING SECTOR: XLK

PROCESSING SECTOR: XLE

--- Granger Causality: Macro -> XLK ---

--- Granger Causality: Macro -> XLE ---

[Unemployment_Surprise]

[Unemployment_Surprise]

Lag 1: p-value = 0.1314

Lag 1: p-value = 0.0033 ***

Lag 2: p-value = 0.4695

Lag 2: p-value = 0.0909 *

Lag 3: p-value = 0.6571

Lag 3: p-value = 0.2894

Lag 4: p-value = 0.5578

Lag 4: p-value = 0.1684

Lag 5: p-value = 0.9265

Lag 5: p-value = 0.1505

[Housing_Starts_Surprise]

[Housing_Starts_Surprise]

Lag 1: p-value = 0.0076 ***

Lag 1: p-value = 0.0096 ***

Lag 2: p-value = 0.0950 *

Lag 2: p-value = 0.1130

Lag 3: p-value = 0.2662

Lag 3: p-value = 0.3199

Lag 4: p-value = 0.5330

Lag 4: p-value = 0.6608

Lag 5: p-value = 0.7472

Lag 5: p-value = 0.8752

PROCESSING SECTOR: XLF

PROCESSING SECTOR: XLV

--- Granger Causality: Macro -> XLF ---

--- Granger Causality: Macro -> XLV ---

[Unemployment_Surprise]

[Unemployment_Surprise]

Lag 1: p-value = 0.0691 *

Lag 1: p-value = 0.2569

Lag 2: p-value = 0.2410

Lag 2: p-value = 0.3493

Lag 3: p-value = 0.6538

Lag 3: p-value = 0.1602

Lag 4: p-value = 0.5822

Lag 4: p-value = 0.1293

Lag 5: p-value = 0.6287

Lag 5: p-value = 0.3272

[Housing_Starts_Surprise]

[Housing_Starts_Surprise]

Lag 1: p-value = 0.0000 ***

Lag 1: p-value = 0.0496 **

Lag 2: p-value = 0.0000 ***

Lag 2: p-value = 0.4402

Lag 3: p-value = 0.0000 ***

Lag 3: p-value = 0.7302

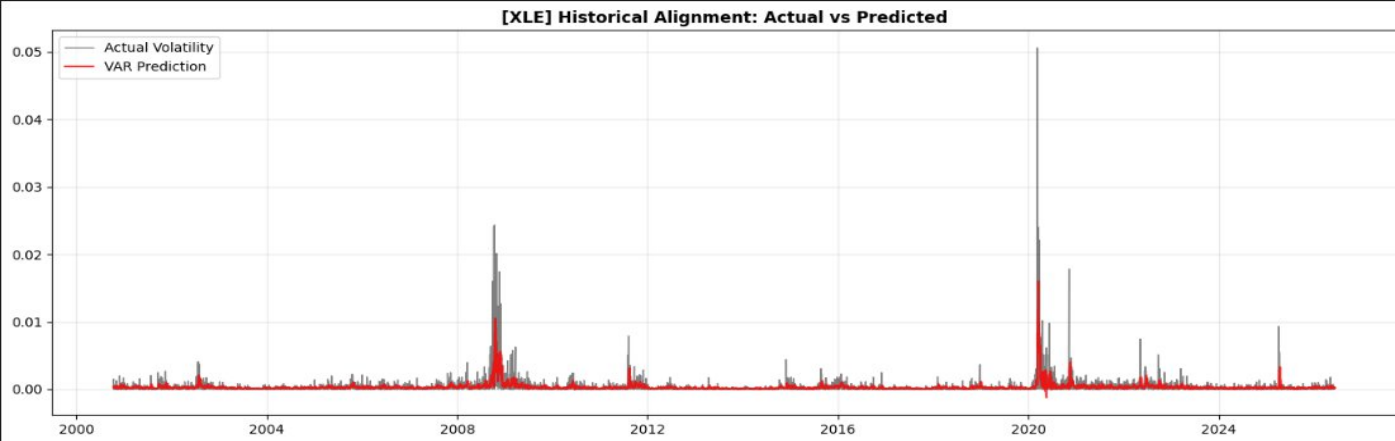
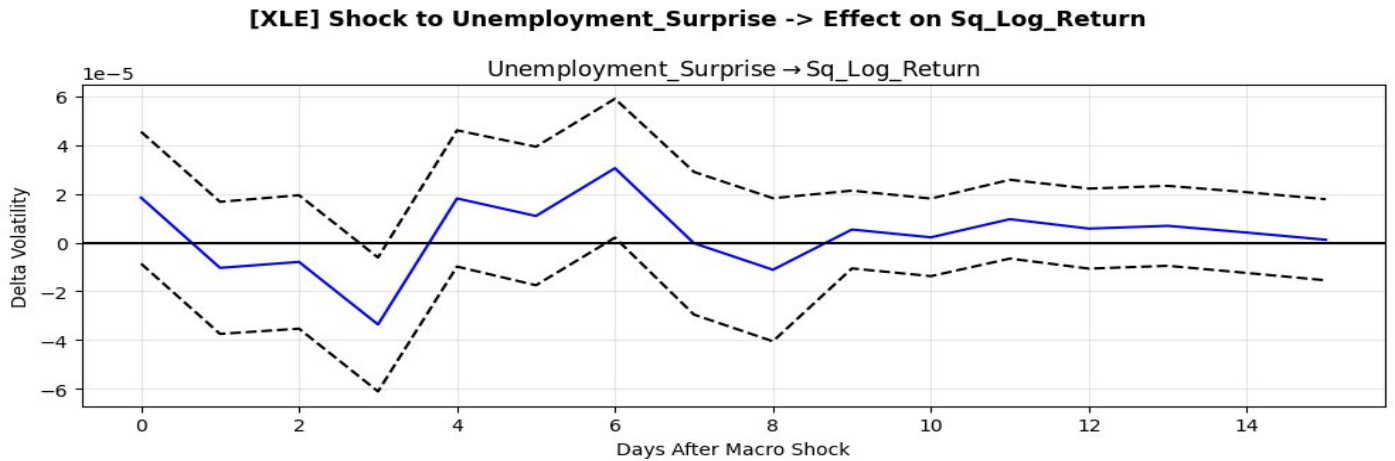
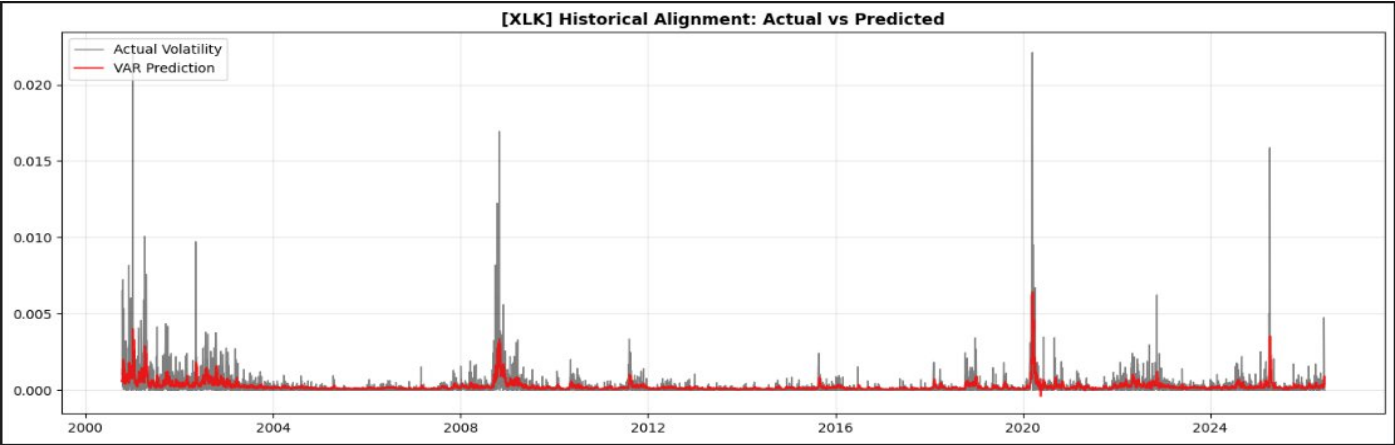
Lag 4: p-value = 0.0003 ***

Lag 4: p-value = 0.8573

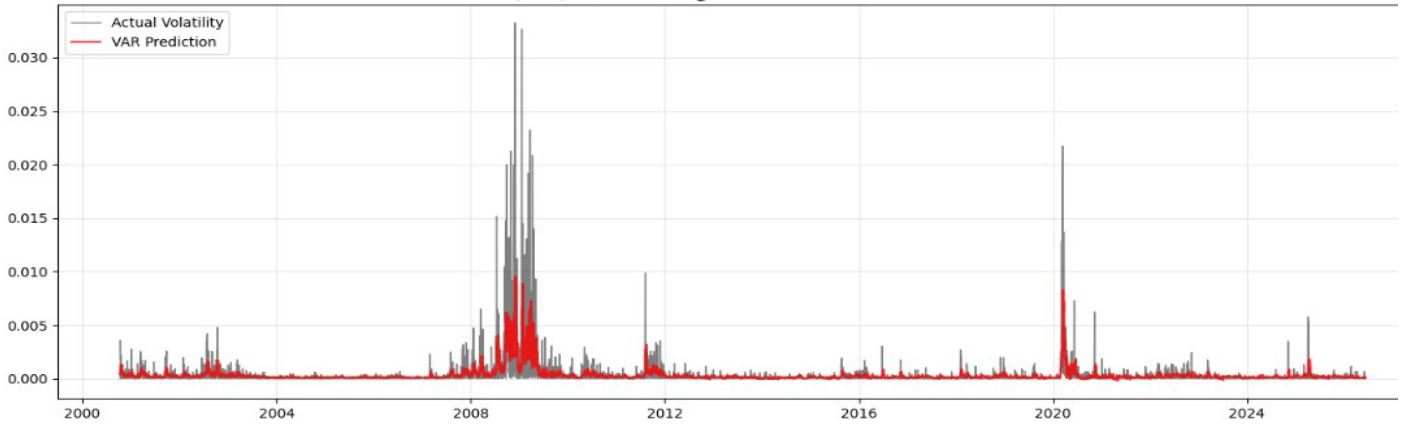
Lag 5: p-value = 0.0035 ***

Lag 5: p-value = 0.9478

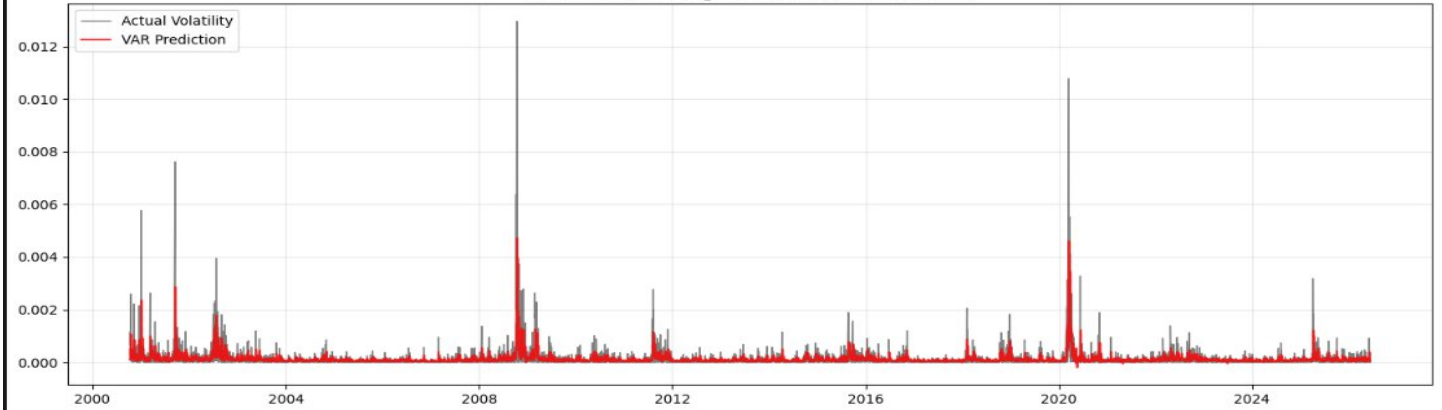
	XLK			XLE			XLF			XLV		
	Coef	P-Val	Sig	Coef	P-Val	Sig	Coef	P-Val	Sig	Coef	P-Val	Sig
const	0.0001	0.0000	***	0.0001	0.0000	***	0.0001	0.0000	***	0.0	0.0000	***
L1.Unemployment_Surprise	-0.0000	0.6667		-0.0001	0.4538		0.0001	0.4586		-0.0	0.0358	**
L3.Unemployment_Surprise	0.0001	0.3653		-0.0001	0.2231		-0.0001	0.5442		0.0	0.0951	*
L4.Unemployment_Surprise	-0.0000	0.4703		0.0002	0.0096	***	0.0001	0.3918		-0.0	0.3278	
L7.Unemployment_Surprise	0.0001	0.3943		-0.0002	0.0965	*	-0.0000	0.8052		-0.0	0.7620	



[XLF] Historical Alignment: Actual vs Predicted



[XLV] Historical Alignment: Actual vs Predicted



macro_shocks = ['CPI_Surprise', 'PPI_Surprise', 'FedFunds_Diff']:

LAG SELECTION CRITERIA

VAR Order Selection (* highlights the minimums)

	AIC	BIC	FPE	HQIC
0	-19.27	-19.27	4.261e-09	-19.27
1	-24.42	-24.40	2.489e-11	-24.41
2	-25.09	-25.05*	1.275e-11	-25.07
3	-25.09	-25.03	1.270e-11	-25.07
4	-25.10	-25.03	1.255e-11	-25.08
5	-25.11	-25.03	1.238e-11	-25.08
6	-25.13	-25.02	1.221e-11	-25.09
7	-25.14	-25.01	1.212e-11	-25.09
8	-25.14	-25.01	1.203e-11	-25.10
9	-25.17*	-25.01	1.177e-11*	-25.11*

PROCESSING SECTOR: XLK

PROCESSING SECTOR: XLE

--- Granger Causality: Macro -> XLK - --- Granger Causality: Macro -> XLE
[CPI_Surprise]

Lag 1: p-value = 0.0215 **
Lag 2: p-value = 0.1791
Lag 3: p-value = 0.4146
Lag 4: p-value = 0.7099
Lag 5: p-value = 0.8378

Lag 1: p-value = 0.0120 **
Lag 2: p-value = 0.0227 **
Lag 3: p-value = 0.0873 *
Lag 4: p-value = 0.3398
Lag 5: p-value = 0.5308

[PPI_Surprise]

Lag 1: p-value = 0.0001 ***
Lag 2: p-value = 0.0075 ***
Lag 3: p-value = 0.0422 **
Lag 4: p-value = 0.1848
Lag 5: p-value = 0.4747

[PPI_Surprise]

Lag 1: p-value = 0.0000 ***
Lag 2: p-value = 0.0000 ***
Lag 3: p-value = 0.0008 ***
Lag 4: p-value = 0.0213 **
Lag 5: p-value = 0.1394

[FedFunds_Diff]

Lag 1: p-value = 0.0180 **
Lag 2: p-value = 0.0000 ***
Lag 3: p-value = 0.0001 ***
Lag 4: p-value = 0.0002 ***
Lag 5: p-value = 0.0002 ***

[FedFunds_Diff]

Lag 1: p-value = 0.9800
Lag 2: p-value = 0.2406
Lag 3: p-value = 0.5517
Lag 4: p-value = 0.3754
Lag 5: p-value = 0.0052 ***

PROCESSING SECTOR: XLF

PROCESSING SECTOR: XLV

--- Granger Causality: Macro -> XLF --- Granger Causality: Macro -> XLV
[CPI_Surprise]

Lag 1: p-value = 0.0000 ***
Lag 2: p-value = 0.0000 ***
Lag 3: p-value = 0.0000 ***
Lag 4: p-value = 0.0000 ***
Lag 5: p-value = 0.0000 ***

[CPI_Surprise]

Lag 1: p-value = 0.0714 *
Lag 2: p-value = 0.4639
Lag 3: p-value = 0.4949
Lag 4: p-value = 0.6491
Lag 5: p-value = 0.8508

[PPI_Surprise]

Lag 1: p-value = 0.0000 ***
Lag 2: p-value = 0.0000 ***
Lag 3: p-value = 0.0000 ***
Lag 4: p-value = 0.0000 ***
Lag 5: p-value = 0.0000 ***

[PPI_Surprise]

Lag 1: p-value = 0.0000 ***
Lag 2: p-value = 0.0419 **
Lag 3: p-value = 0.0670 *
Lag 4: p-value = 0.1187
Lag 5: p-value = 0.3079

[FedFunds_Diff]

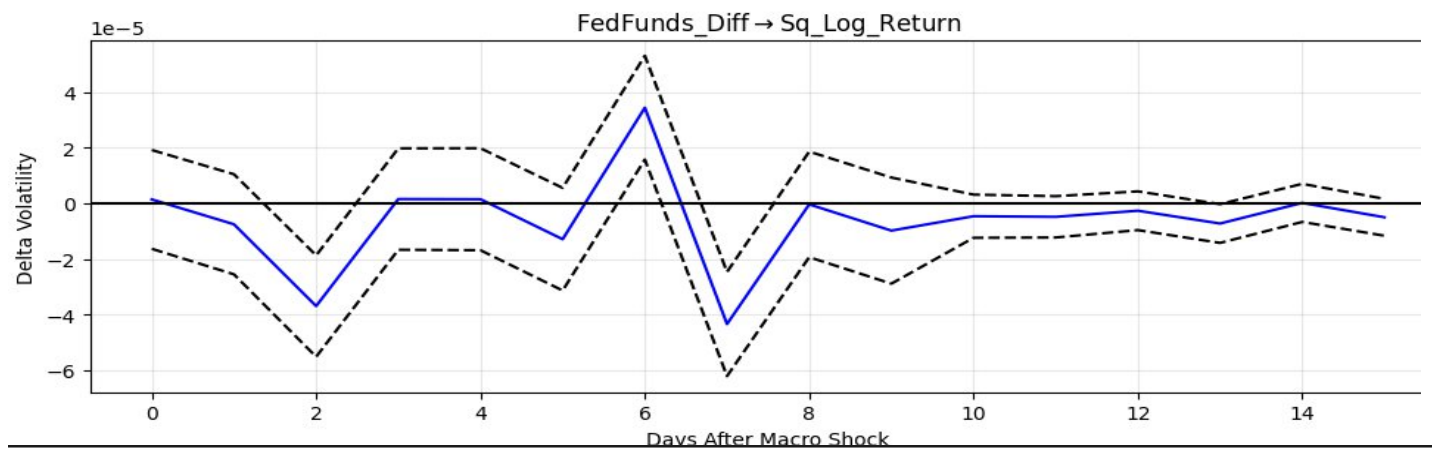
Lag 1: p-value = 0.3614
Lag 2: p-value = 0.1931
Lag 3: p-value = 0.0509 *
Lag 4: p-value = 0.4027
Lag 5: p-value = 0.0914 *

[FedFunds_Diff]

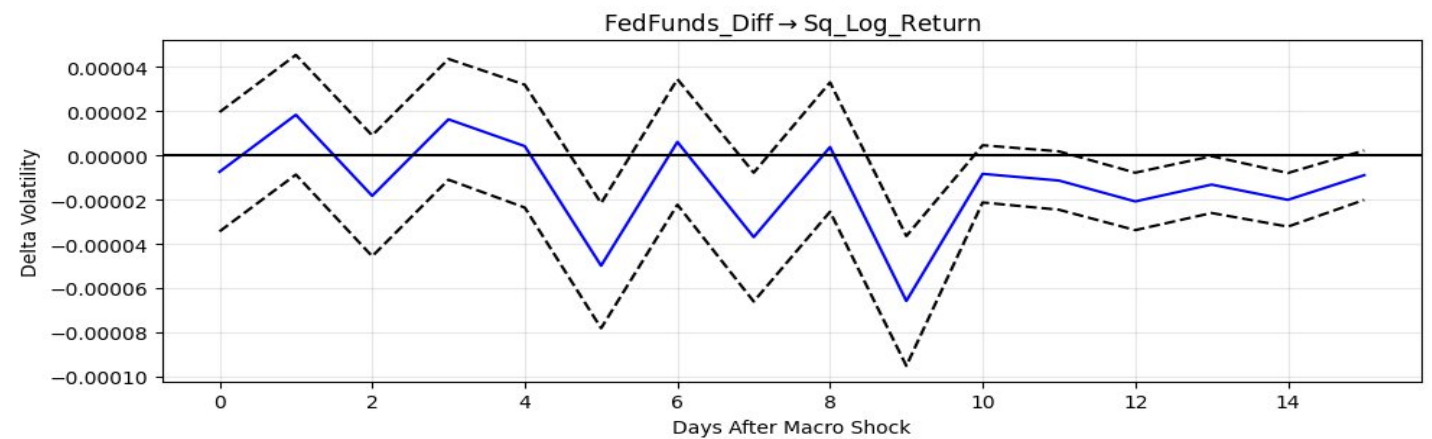
Lag 1: p-value = 0.1175
Lag 2: p-value = 0.0009 ***
Lag 3: p-value = 0.0055 ***
Lag 4: p-value = 0.0136 **
Lag 5: p-value = 0.0001 ***

	XLK			XLE			XLF			XLV		
	Coef	P-Val	Slg	Coef	P-Val	Slg	Coef	P-Val	Slg	Coef	P-Val	Slg
const	0.0001	0.0000	***	0.0001	0.0000	***	0.0001	0.0000	***	0.0000	0.0000	***
L1.CPI_Surprise	-0.0000	0.5762		0.0001	0.3613		0.0002	0.0134	**	-0.0000	0.7977	
L2.CPI_Surprise	0.0000	0.6447		-0.0002	0.1205		-0.0002	0.0986	*	0.0000	0.3252	
L2.FedFunds_Diff	-0.0007	0.0001	***	-0.0003	0.2186		-0.0001	0.6068		-0.0003	0.0009	***
L5.FedFunds_Diff	-0.0002	0.1774		-0.0010	0.0001	***	-0.0004	0.2089		-0.0003	0.0002	***
L6.FedFunds_Diff	0.0008	0.0000	***	0.0001	0.7778		0.0002	0.5663		0.0000	0.8991	
L7.FedFunds_Diff	-0.0008	0.0000	***	-0.0006	0.0339	**	-0.0001	0.6911		-0.0006	0.0000	***
L8.FedFunds_Diff	0.0001	0.6211		0.0000	0.9102		0.0001	0.6407		0.0002	0.0349	**
L9.FedFunds_Diff	-0.0001	0.7438		-0.0010	0.0002	***	-0.0003	0.3113		-0.0003	0.0001	***

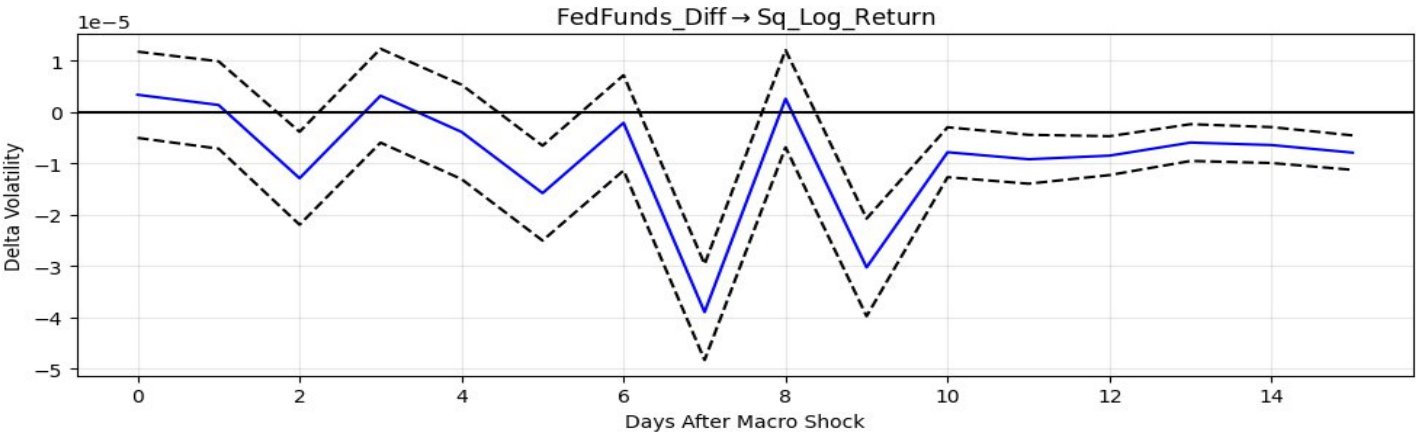
[XLK] Shock to FedFunds_Diff -> Effect on Sq_Log_Return



[XLE] Shock to FedFunds_Diff -> Effect on Sq_Log_Return



[XLV] Shock to FedFunds_Diff -> Effect on Sq_Log_Return



macro_shocks = ['Trade_Balance_Surprise', 'M1_Money_Surprise']:

LAG SELECTION CRITERIA				
VAR Order Selection (* highlights the minimums)				
	AIC	BIC	FPE	HQIC
0	17.25	17.26	3.115e+07	17.26
1	12.24	12.25	2.059e+05	12.24
2	12.19	12.21	1.961e+05	12.19
3	12.18	12.21	1.949e+05	12.19
4	12.16	12.21	1.919e+05	12.18
5	12.15	12.20	1.897e+05	12.17
6	12.13	12.19*	1.861e+05	12.15
7	12.13	12.20	1.862e+05	12.16
8	12.12	12.20	1.840e+05	12.15*
9	12.12*	12.21	1.837e+05*	12.15

PROCESSING SECTOR: XLK

PROCESSING SECTOR: XLE

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--- Granger Causality: Macro -> XLK --- Granger Causality: Macro -> XLE
[Trade_Balance_Surprise]
Lag 1: p-value = 0.0556 *
Lag 2: p-value = 0.3292
Lag 3: p-value = 0.6210
Lag 4: p-value = 0.7146
Lag 5: p-value = 0.8590
[M1_Money_Surprise]
Lag 1: p-value = 0.5513
Lag 2: p-value = 0.8849
Lag 3: p-value = 0.9532
Lag 4: p-value = 0.9868
Lag 5: p-value = 0.9962

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PROCESSING SECTOR: XLF

PROCESSING SECTOR: XLV

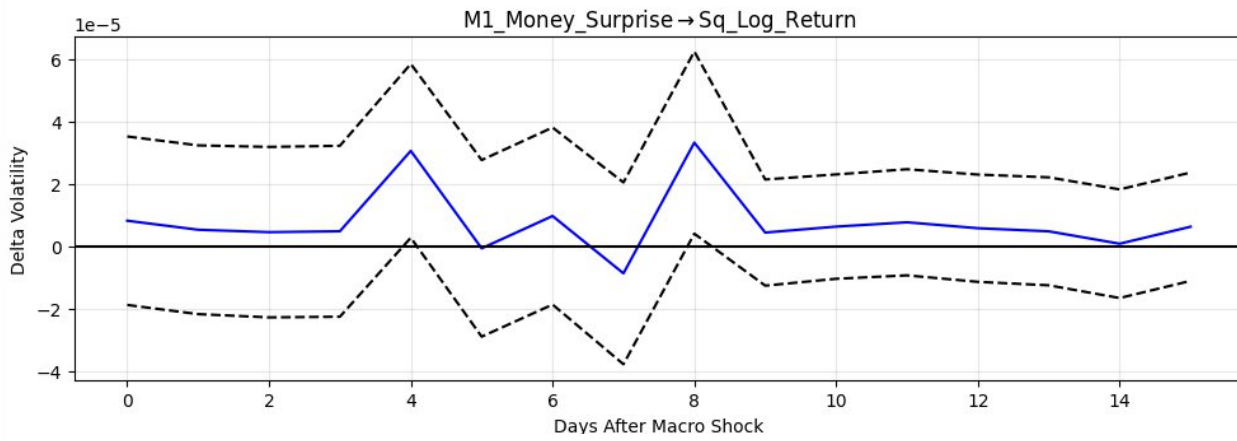
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--- Granger Causality: Macro -> XLF --- Granger Causality: Macro -> XLV ---
[Trade_Balance_Surprise]
Lag 1: p-value = 0.0000 ***
Lag 2: p-value = 0.0000 ***
Lag 3: p-value = 0.0000 ***
Lag 4: p-value = 0.0008 ***
Lag 5: p-value = 0.0057 ***
[M1_Money_Surprise]
Lag 1: p-value = 0.0941 *
Lag 2: p-value = 0.3651
Lag 3: p-value = 0.6331
Lag 4: p-value = 0.8289
Lag 5: p-value = 0.9322

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	XLK			XLE			XLF			XLV		
	Coef	P-Val	Sig	Coef	P-Val	Sig	Coef	P-Val	Sig	Coef	P-Val	Sig
const	0.0001	0.0000	***	0.0001	0.0000	***	0.0001	0.0000	***	0.0	0.0000	***
L3.Trade_Balance_Surprise	0.0000	0.5584		0.0000	0.3744		0.0000	0.4828		0.0	0.0868	*
L8.Trade_Balance_Surprise	-0.0000	0.9611		0.0000	0.1967		0.0000	0.0141	**	0.0	0.9325	
L8.M1_Money_Surprise	0.0000	0.2094		0.0000	0.0374	**	0.0000	0.3526		0.0	0.0129	**
L9.Trade_Balance_Surprise	0.0000	0.9349		-0.0000	0.7377		-0.0000	0.0018	***	-0.0	0.8324	
L9.M1_Money_Surprise	-0.0000	0.0558	*	-0.0000	0.0643	*	-0.0000	0.1919		-0.0	0.0009	***

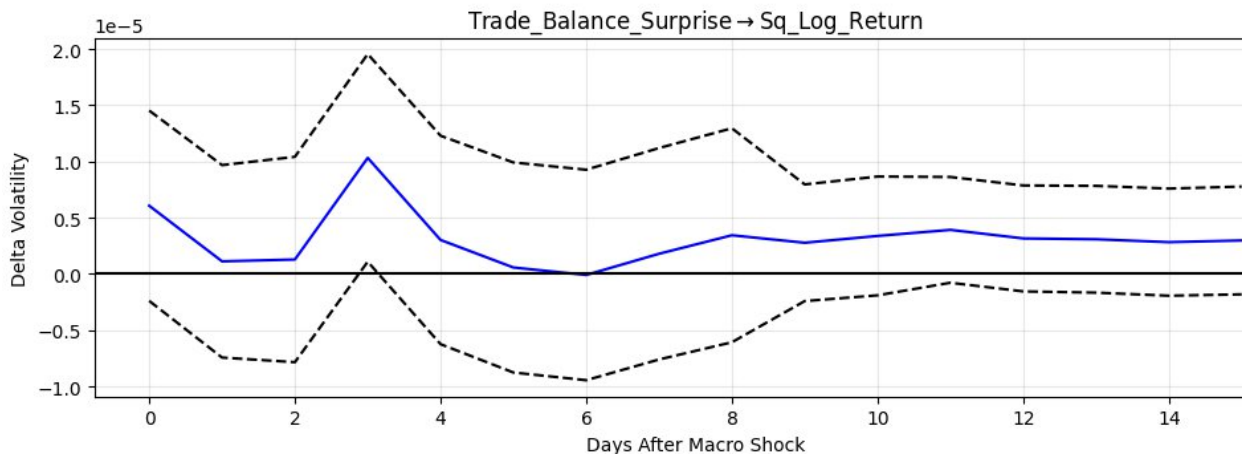
[XLE] Shock to M1_Money_Surprise -> Effect on Sq_Log_Return



[XLF] Shock to Trade_Balance_Surprise -> Effect on Sq_Log_Return



[XLV] Shock to Trade_Balance_Surprise -> Effect on Sq_Log_Return



[XLV] Shock to M1_Money_Surprise -> Effect on Sq_Log_Return

